PORTFOLIO

Ahmed Tharwat Muhammad Ibrahim

AI & Data Science | Machine Learning | Data Analysis

"Turning Data Into Intelligent Solutions."

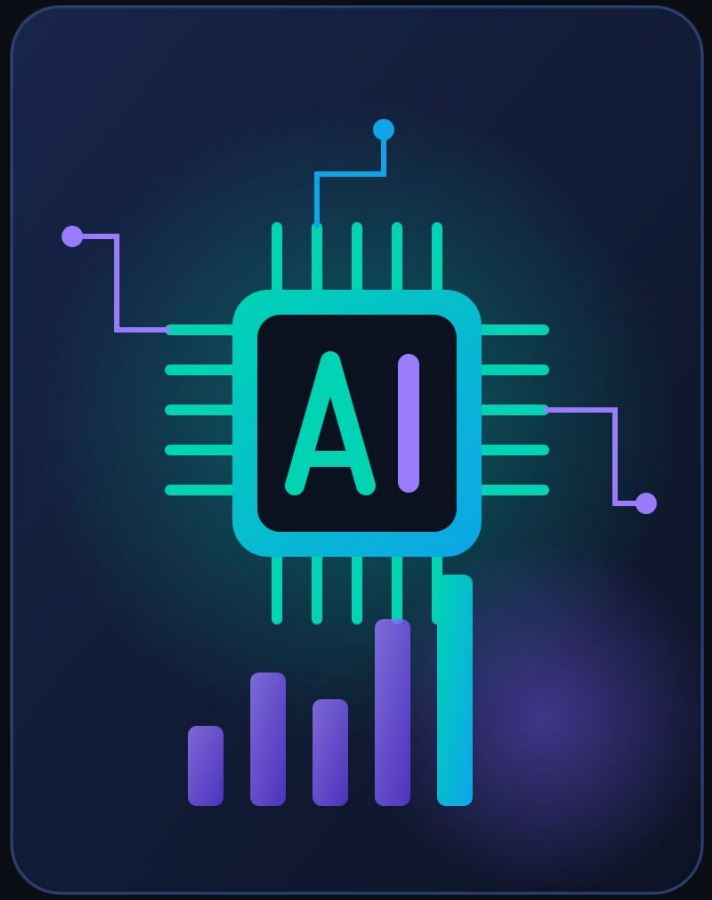
Machine Learning

Data Analysis

Artificial Intelligence



About Me



01

I specialize in Machine Learning and AI solutions, combining analytical thinking, problem-solving skills, and a deep passion for transforming raw data into meaningful insights.

02

I have a strong ability to analyze complex challenges, identify hidden patterns, and develop intelligent, data-driven models that deliver accurate, efficient, and impactful real-world solutions.

03

My approach focuses not only on building technically strong systems, but also on creating practical AI applications that enhance decision-making, improve performance, and solve real business problems effectively.

04

I am continuously improving my skills in Machine Learning, Data Analysis, and Artificial Intelligence through real-world projects and hands-on experience.

Education & Training

Until 2028

Zagazig National University

Faculty of Computers and Information

Artificial Intelligence & Data Analysis Department

In Progress

Digital Egypt Pioneers Initiative (DEPI)

AI & Data Science: Data Scientist Track (IBM)

- Python, SQL and Data Visualization
- Machine Learning with Python
- MLOps: MLflow and Hugging Face

Until 2026

National Telecommunication Institute (NTI)

Machine Learning Training Program

- Machine Learning Fundamentals
- Data Analysis & Modeling
- Real-world ML Projects

Featured Project

Bank Term Deposit Prediction — ML Pipeline

Role: Data Scientist / ML Engineer • Duration: 1 Week • Status: Completed



Overview

End-to-end ML pipeline predicting bank term deposit subscriptions using UCI Bank Marketing Dataset (~45K records). Demonstrates real-world ML workflow from data processing to model evaluation.

Key Challenges

- Severe class imbalance (~88% vs ~12%)
- Avoiding data leakage
- Ensuring reproducibility
- Improving minority class prediction

Model Results

BEST MODEL

- XGBoost — 82.4%
- Random Forest — 81.6%
- Logistic Regression — 76.3%

Technologies

Python | pandas | NumPy | scikit-learn | matplotlib | seaborn | XGBoost

Featured Project

Sales Analysis Report — Python & Excel

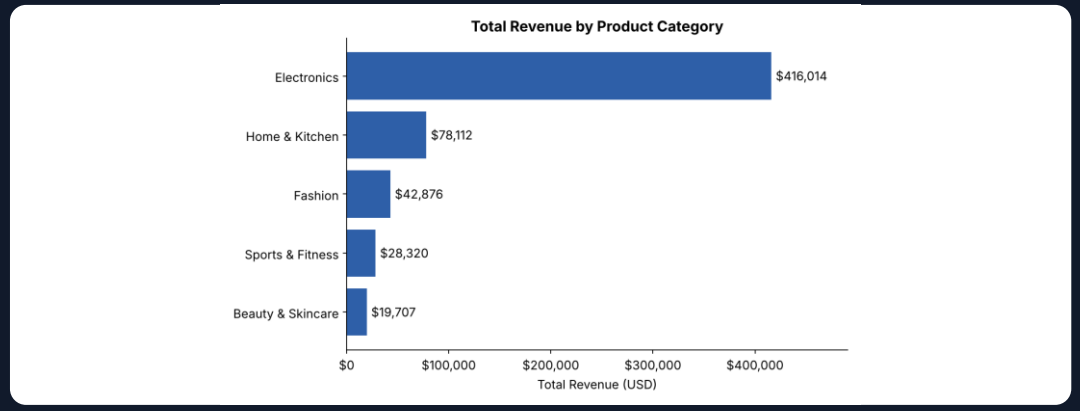
Data Analysis • Python & Excel • Status: Completed

Overview

Python analysis of 500 e-commerce orders (Jan 2025 to Jun 2026): data cleaning, EDA, six charts, business insights and an Excel report generated with openpyxl.

Key Findings

- $585K total revenue from 500 orders
- Electronics drives 71% of revenue
- Riyadh and Jeddah bring 69% of revenue
- Mobile App is the top channel (43.6%)



Technologies

Python | pandas | NumPy | Matplotlib | openpyxl | Excel

Featured Project

E-commerce Sales Analysis

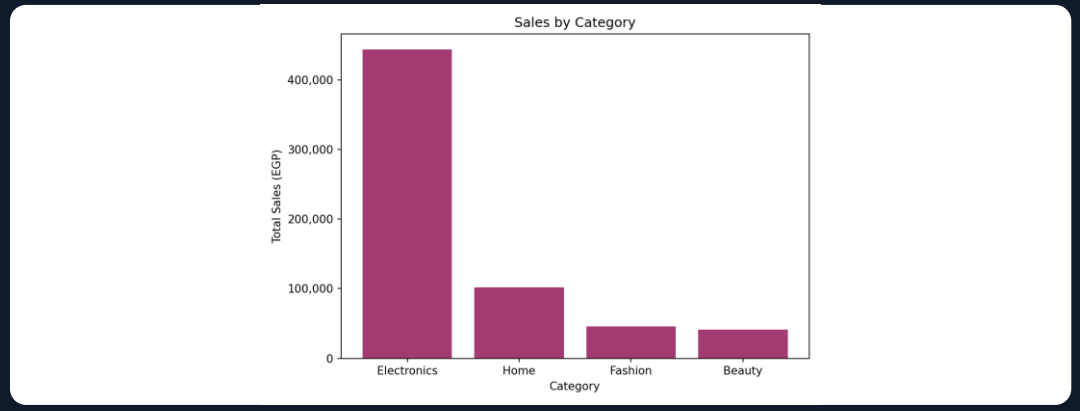
Data Analysis • Python & Excel • Status: Completed

Overview

Analysis of an e-commerce sales dataset (75 orders): data cleaning, EDA, six charts, business insights and an Excel report with live formulas.

Key Findings

- 631,400 EGP in completed-order sales
- 97.3% order completion rate
- Electronics is about 70% of sales
- Laptop is the top product, Cairo the top city



Technologies

Python | pandas | NumPy | Matplotlib | openpyxl | Excel

github.com/Ahmedtharwatt111/Ecommerce_Sales_Analysis

Featured Project

Loan Approval Prediction

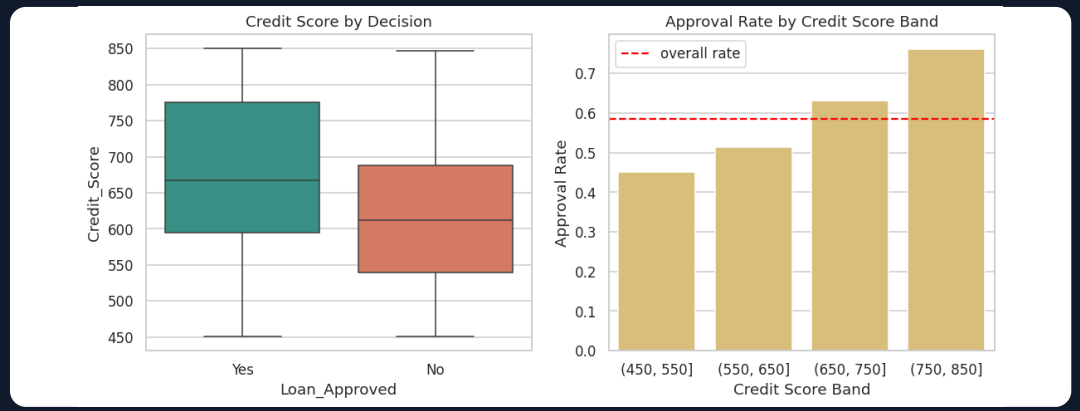
Machine Learning • Classification • Status: Completed

Overview

EDA and a Logistic Regression model that predicts loan approval from 500 applications, with feature engineering, cross-validated regularization and a full evaluation.

Key Findings

- Credit Score is the only strong driver: approval rises from 45% to 76%
- 61% accuracy vs 58.6% baseline (ROC-AUC 0.62)
- Recall 0.73 is higher than precision 0.65



Technologies

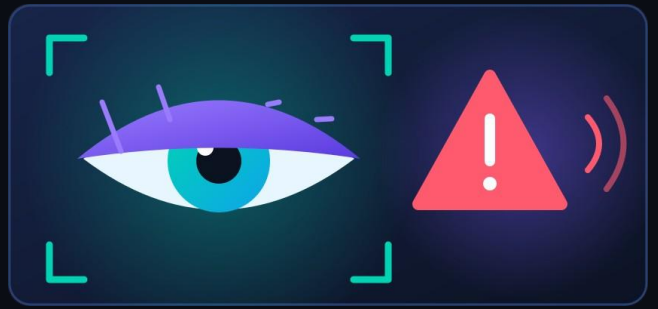
Python | pandas | NumPy | scikit-learn | Matplotlib | seaborn

github.com/Ahmedtharwatt111/Loan_Approval_Prediction

Featured Project

DrowsyGuard AI — Driver Drowsiness Detection

Team Project • Machine Learning & Computer Vision • Status: In Progress



Overview

A team project that uses machine learning and computer vision to detect driver drowsiness in real time and trigger an alert before fatigue leads to an accident.

Key Challenges

- Real-time detection speed
- Varying lighting and driver conditions
- Reducing false alarms
- Balancing accuracy and latency

How It Works

1. Capture driver face via camera
2. Detect drowsiness signs with ML
3. Trigger real-time alert

Technologies

Python | Machine Learning | Computer Vision | Real-time Alerts

Skills & Tools

Skills

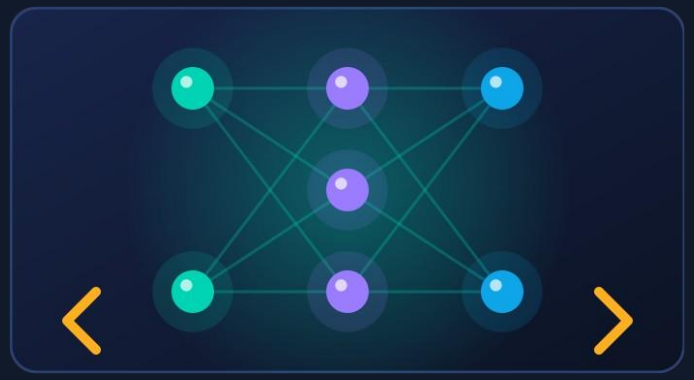
- Machine Learning Model Development
- Data Analysis & Data Preprocessing
- Model Evaluation & Optimization
- Problem Solving using AI
- End-to-End ML Pipeline Development
- SQL & MLOps with MLflow



Tools

VS Code Python Jupyter GitHub

Offered Services



ML Model Development

Custom machine learning models tailored to your business needs — from classification and regression to clustering and recommendation systems.



Data Analysis Consulting

Transform raw data into actionable insights. Comprehensive data exploration, visualization, and statistical analysis to drive informed decisions.



AI Solution Prototyping

Rapid prototyping of AI-powered solutions. From concept to working prototype, validating ideas quickly and cost-effectively.

Let's Connect

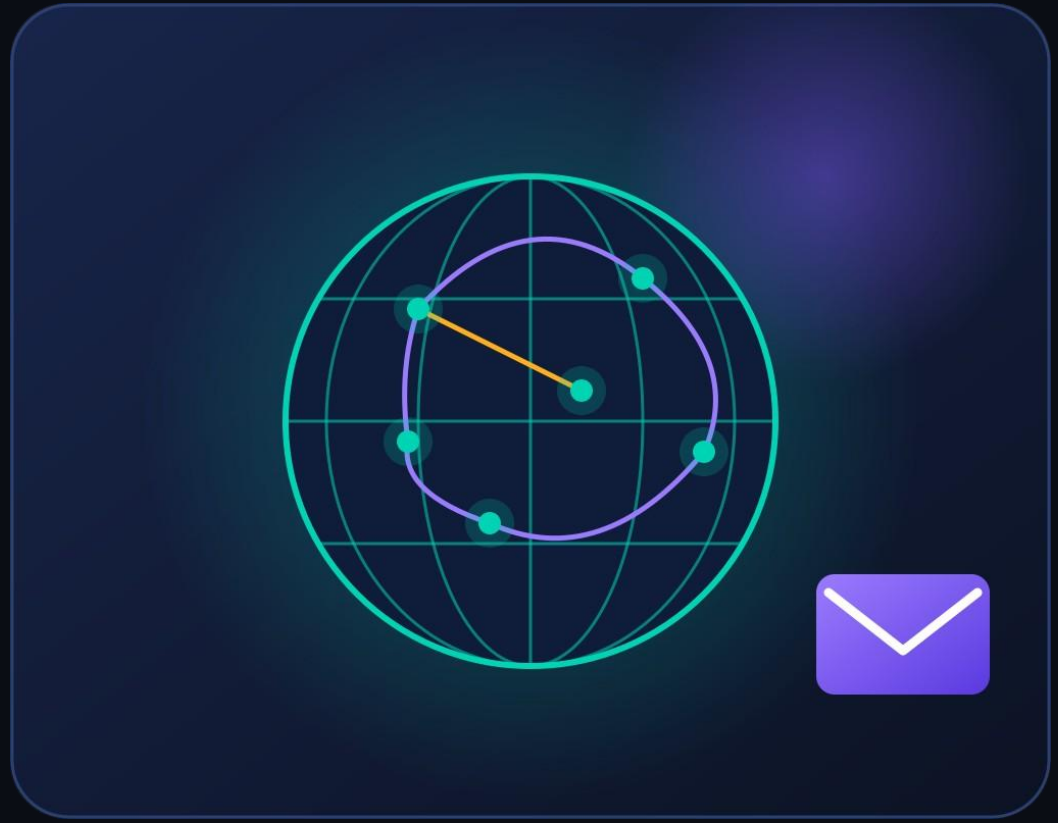
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